

ASSIGNMENT 1

G O M O L E M O
S I B I Y A

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Introduction

Virtual Reality (VR) immerses users directly inside digital environments, making user experience design fundamentally different from traditional 2D interfaces. Poor choices can quickly lead to physical discomfort, motion sickness, disorientation, or even abandonment of the experience.

The core goal is simple yet critical: **comfort comes first**. Every decision must prioritize the user's physiological and psychological well-being before pursuing immersion, engagement, or visual spectacle.

These guidelines bring together widely respected best practices from leading VR platforms (Meta Horizon OS, Unity, Unreal Engine), industry research, developer communities, and real-world lessons learned. They are structured to guide you step by step:

- **Foundational Principles** (user focus, comfort as priority, core physiological and ergonomic basics)
- **Spatial and Immersion Design** (scale, environment, presence-building elements)
- **Interaction and Input Design** (natural controls, feedback, UI presentation)
- **Movement and Navigation** (locomotion choices, user agency)
- **Performance, Technical, and Accessibility Essentials** (optimization, inclusivity, safety)
- **Onboarding, Testing, and Ethical Practices** (learnability, validation, broader considerations)

By following these principles, you can craft VR/MR experiences that feel natural and trustworthy, support longer comfortable sessions, welcome users of all abilities, and foster true “being there” presence without compromising health, trust, or inclusivity.

Foundational Principles: User-Centred Design and Human Comfort

These form the bedrock of VR UX, prioritizing empathy, physiological safety, and real-user alignment to prevent frustration, fatigue, and cybersickness.

- **Prioritize user needs and empathy** Conduct thorough user research (interviews, surveys, observational studies, card sorting) to understand needs, expectations, pain points, and diverse contexts. Design with empathy for physical and emotional experiences in VR to align with real behaviours and reduce frustration.
- **Respect handedness and inclusivity** Honour system-level left- or right-handed preferences for menus and primary interactions to promote natural use and prevent discomfort.
- **Ensure physiological comfort and prevent motion sickness** Maintain high frame rates (≥ 90 FPS ideally, minimum 60 FPS) and low latency (< 20 ms motion-to-photon). Avoid forced camera movement, rapid acceleration, shaking, or unexpected rotations. Offer user-controlled options like teleportation, guided locomotion, snap turning (discrete increments), vignettes/tunnelling, and seated modes. Provide stable horizon lines, visual reference points, and fixed elements (e.g., dashboards) to minimize sensory conflicts.
- **Providing User-Controlled Locomotion Options**
Always offer multiple user-selectable locomotion methods to match individual comfort preferences, including:
 - Teleportation
 - Guided (smooth) locomotion
 - Snap turning (discrete increments instead of continuous rotation)
 - Tunnelling during movement
 - Seated mode support
- **Design for realistic human scale and ergonomics** Use 1:1 real-world scale for all objects, environments, and characters to avoid subconscious detection of inconsistencies that break immersion or cause disorientation. Position content at comfortable distances (roughly 1 meter, slightly below eye level to start), within optimal zones (0.5–10 meters, 94° horizontal/ 32° vertical field of view). Avoid spawning objects too close/large or requiring excessive head/neck movement.
- **Support session length and natural breaks** Structure experiences for 20–40-minute core loops to match headset limitations (weight, fit). Include natural break points every 10–15 minutes for pausing/exiting without

progress loss, plus session-time reminders and low-exit-cost features (auto-saves, clear objectives).

- **Promote physical and virtual safety** Implement boundary/guardian systems, respect personal space (avoid unexpected proximity of characters), protect against real-world hazards, ensure privacy/identity control, and design inclusively for varying abilities.

Prototyping, Onboarding, and Early Engagement

Early validation and intuitive starts are critical in VR, where issues only surface in-headset.

- **Prototype early and often with in-headset testing** Start with simple MVPs (Unity, Unreal Engine, A-Frame, Google Blocks); sketch on paper first. Conduct usability testing in-headset (not desktop) to measure sickness, fatigue, disorientation, scale, and interactions. Iterate frequently based on real feedback—what works on 2D screens often fails in VR.
- **Create effective onboarding and learnability** Use interactive, in-VR tutorials where users physically perform actions (reach, grab, throw) to build muscle memory. Introduce mechanics gradually via contextual cues and environmental storytelling. Avoid lengthy non-interactive cinematics.
- **Shorten time to fun and initial engagement** Engage users immediately with a strong first 30 seconds, quick sense of achievement/progression, and minimal setup hurdles to boost retention.

Spatial Design, Immersion, and Environmental Principles

Focus on creating believable, presence-enhancing 3D worlds that feel natural and grounded.

- **Leverage spatial awareness and environmental understanding** Use real-world scale, consistent gravity/lighting/physics, natural affordances (usable-looking doors/buttons), landmarks for orientation, and uncluttered spaces. Align virtual objects with physical surfaces (in mixed reality) with shadows for grounded realism.
- **Enhance immersion through multisensory and aesthetic design** Build natural environments with thoughtful colours, lighting, sounds, and spatial audio (3D, position/orientation-relative) to deepen presence. Use environmental storytelling (light, pathing, audio, interesting objects) for subtle guidance instead of intrusive prompts.
- **Maintain visual consistency and readability** Use angular scaling for UI elements to keep constant perceived size/readability at any distance. Apply billboard (gentle facing rotation) for critical panels/labels without

feeling unnatural. Ensure minimum angular text size (e.g., 1.5° FOV) and high contrast.

Interaction and Input Design

Interactions should feel physical, intuitive, and effortless in 3D space.

- **Leverage physical-world metaphors and natural controls** Base actions on real-life behaviours (grab, press, pull) for zero learning curve. Support multimodal inputs (head gestures, hand/arm movements, voice, eye tracking, body) for flexibility and accessibility.
- **Support direct, indirect, and flexible interactions** Allow nearby direct grabbing plus distant ray-casting/pointing (with consistent arm model, slight downward angle ~15°). Balance two-handed use, avoid precise micro-movements (to reduce fatigue/frustration), and make actions discoverable.
- **Provide rich, multi-modal feedback** Combine visual (glows, highlights, compression motion), audio (3D effects, confirmations), and haptic (vibrations for contact/grab/release) cues for every action to compensate for lack of real tactile feedback and reinforce safety/boundaries.
- **Optimize UI presentation and hierarchy** Embed diegetic UI (world-integrated panels, watches, tools) instead of traditional 2D/head-locked (unless essential). Maintain consistency/hierarchy via size, contrast, colour, placement. Minimize text; prioritize icons, visual cues, and gradual introductions.

Performance, Control, and Agency

Technical reliability and user empowerment preserve immersion and trust.

- **Ensure high performance and technical stability** Optimise assets (occlusion culling, LOD), use seamless transitions, reduce loading interruptions, and handle tracking loss gracefully. Maintain stable high frame rates to prevent judder/discomfort.
- **Prioritize user agency, control, and psychological safety** Allow instant pause, exit, recentre. Avoid trapping users, manipulative stimuli without warning, or deceptive elements. Provide clear escape options and boundaries between simulation/reality.
- **Offer flexible locomotion and movement** Never force one method—provide teleportation (comfort default), smooth locomotion (with vignetting), room-scale, walking-in-place. Use fade transitions and align with real-world expectations.

Evaluation, Ethics, and Futureproofing

Close the loop with testing and responsible design.

- **Conduct rigorous evaluation and user testing** Perform in-headset testing with real users; observe movement, ergonomics, sickness, presence, and comfort. Gather qualitative/quantitative feedback and iterate frequently.
- **Address ethical considerations** Protect biometric/movement data, avoid addictive/manipulative loops, ensure transparency in recording, prioritize wellbeing over metrics, promote honesty/freedom, and give control over choices/data.
- **Design for accessibility and all users** Offer customization (text size/colour, feedback types) for impairments. Adapt to diverse physicalities with simple navigation.

Incorporate emerging trends and standards Stay updated on haptic gloves, advanced hand tracking, AI personalization, brain-computer interfaces. Follow platform guidelines (e.g., Meta, Google) and emerging standards (e.g., ISO/IEC for AR/MR/VR UX).

Comprehensive VR UX design checklist

1. Foundational Principles: User-Centred Design and Human Comfort

- ☐ Conduct thorough user research (interviews, surveys, observations, card sorting) to map needs, expectations, pain points, and diverse contexts
- ☐ Design with empathy for physical/emotional VR experiences (consider fatigue, disorientation, emotional impact)
- ☐ Respect system-level handedness preferences for controls, menus, and primary interactions
- ☐ Offer customization for accessibility (text size/colour/contrast, feedback intensity, input alternatives)
- ☐ Design inclusively for varying physical abilities, motor skills, visual/auditory impairments, and age ranges
- ☐ Include options for seated vs. standing/room-scale use
- ☐ Provide clear privacy controls for movement/biometric data tracking

2. Physiological Comfort and Motion Sickness Prevention

- ☐ Target ≥ 90 FPS (minimum stable 60–72 FPS) with low motion-to-photon latency (< 20 ms)
- ☐ Avoid forced/ unexpected camera movement, rapid acceleration/deceleration, shaking, or abrupt rotations
- ☐ Offer user-controlled locomotion defaults: teleportation as primary comfort option
- ☐ Include comfort settings: snap turning (15 – 30° increments), vignettes/tunnelling during motion, seated/standing modes, brightness/contrast adjustments
- ☐ Maintain a stable, level horizon line and fixed visual reference points (e.g., dashboards, ground textures)
- ☐ Limit session design to 20–40 minute core loops; build in natural break points every 10–15 minutes
- ☐ Add session timers, fatigue reminders, and easy pause/exit without progress loss
- ☐ Use angular scaling for UI to maintain consistent perceived size/legibility at varying distances

- ☐ Ensure minimum angular text size ($\sim 1.5^\circ$ FOV) with high contrast, anti-aliasing, and readable fonts
- ☐ Position critical content in comfortable viewing zones (0.5–10 m distance, within $\sim 94^\circ$ horizontal / 32° vertical comfortable FOV)

3. Prototyping, Testing, and Iteration

- ☐ Prototype early/often using tools like Unity/Unreal/A-Frame/Google Blocks; start with paper sketches
- ☐ Conduct frequent in-headset usability testing (never rely solely on desktop simulation)
- ☐ Measure real metrics: cybersickness, neck/eye strain, fatigue, disorientation, presence, task completion
- ☐ Observe physical ergonomics (posture, movement range, gorilla-arm fatigue for hand tracking)
- ☐ Iterate based on qualitative feedback (comfort, intuitiveness, immersion) and quantitative data
- ☐ Test rough prototypes as early as possible—what works in 2D often fails in 3D

4. Onboarding, Learnability, and Initial Engagement

- ☐ Create interactive, hands-on in-VR tutorials (users physically perform actions: reach/grab/throw)
- ☐ Introduce mechanics gradually via contextual/environmental cues rather than long text/instructions
- ☐ Use environmental storytelling (lighting, paths, spatial audio, landmarks) for subtle guidance
- ☐ Minimize non-interactive intros; achieve "time to fun" with quick achievement/progression in first 1–3 minutes
- ☐ Provide strong first 30 seconds to build immediate presence and reduce setup hurdles

5. Spatial Design, Immersion, and Environmental Grounding

- ☐ Enforce realistic 1:1 real-world scale for objects, environments, characters to preserve immersion
- ☐ Align virtual objects with physical surfaces (mixed reality) using perfect placement and shadows
- ☐ Use natural affordances (doors/buttons look pressable/grabbable) and consistent physics/gravity/lighting

- ☐ Provide landmarks, uncluttered spaces, and clear spatial orientation cues to prevent disorientation
- ☐ Implement 3D spatial audio (position/orientation-relative) for immersion, attention guidance, and cues
- ☐ Leverage thoughtful aesthetics (colours, lighting, sounds) to enhance mood/engagement without strain
- ☐ Avoid visual overload; use subtle cues over intrusive arrows/text prompts

6. Interaction and Input Design

- ☐ Base interactions on real-world physical metaphors (grab/press/pull) for intuitive zero-learning-curve
- ☐ Support multimodal inputs: hand gestures, controllers, voice commands, eye tracking, head/body movements
- ☐ Enable direct manipulation (nearby grab) + indirect (ray-casting/pointing with ~15° downward arm model)
- ☐ Support two-handed interactions where natural; avoid requiring precise micro-movements (fatigue risk)
- ☐ Make actions discoverable (visual hints, progressive disclosure) without mandatory tutorials
- ☐ Provide rich multi-modal feedback: visual (glows/highlights/compression), audio (clicks/whooshes/3D confirmations), haptic (vibrations for contact/grab/release/texture simulation)
- ☐ Use billboarding for critical panels/labels (gentle facing rotation, not locked to head)
- ☐ Ensure large, spaced hit targets for hand tracking/voice/gaze (avoid small/precise requirements)

7. UI Presentation and Information Design

- ☐ Prefer diegetic/world-embedded UI (panels as objects/tools/watches) over flat/head-locked 2D
- ☐ Minimize text; prioritize icons, visual metaphors, spatial cues, and gradual introductions
- ☐ Establish clear visual hierarchy (size/contrast/colour/placement) consistent across 360° views
- ☐ Avoid clutter; show only essential info at any moment to reduce cognitive load

8. Performance, Technical Reliability, and Agency

- ☐ Optimize for stable high frame rates (occlusion culling, LOD, efficient assets)
- ☐ Use seamless transitions; minimize loading interruptions or overheating risks
- ☐ Handle tracking loss gracefully with fallback behaviour and clear user guidance
- ☐ Give full user agency: instant pause/recentre/exit, no trapping experiences
- ☐ Respect psychological safety: avoid unexpected proximity, distressing stimuli without consent/warning
- ☐ Offer multiple locomotion options (teleport default + smooth with vignette, room-scale, walking-in-place)
- ☐ Use fade transitions instead of abrupt scene changes

9. Ethical, Accessibility, and Future-Proof Considerations

- ☐ Protect user data (biometrics, movements); ensure transparency and opt-in controls
- ☐ Avoid addictive/manipulative loops; prioritize wellbeing over pure engagement metrics
- ☐ Promote honesty: no deception/coercion; clear boundaries between virtual/real
- ☐ Adhere to platform standards (Meta Horizon OS, Google, emerging ISO/IEC for XR)
- ☐ Prepare for emerging inputs (advanced haptics, full hand tracking, AI personalization, BCI)
- ☐ Design for mixed reality continuity (persistent anchors, physical-virtual blending)

10. Evaluation and Validation

- ☐ Perform mixed-method testing: in-headset observation + post-session surveys/interviews
- ☐ Collect feedback on presence, comfort, usability, enjoyment, and any nausea/strain
- ☐ Iterate frequently throughout development based on real-user data

02 THREE DIFFERENT GENERATIVE AI TOOLS TO BUILD A COMPREHENSIVE CHECKLIST.

High-Level Overview Table

In this comparison, I evaluate Grok (xAI), ChatGPT, and DeepSeek based on the VR UX design checklist each produces. The side-by-side look will point out what they have in common, where they clearly split apart, what each model does especially well or noticeably poorly, and the particular angle or flavour that makes each one useful for this job.

<i>Aspect</i>	<i>Grok (xAI)</i>	<i>ChatGPT</i>	<i>Deepseek</i>
<i>Strength</i>	8 main categories Principle-based	12 sections with checkboxes Very practical	7 themed pillars Detailed sub-bullets
<i>Readability/Format</i>	Clean paragraphs + bullets	Checkbox list	Hierarchical bullets with bold explanations
<i>Length/Depth</i>	Balanced – concise yet complete	Quite detailed + extra notes & tips	Deep & technical
<i>Best Suited For</i>	Designers	Students, assignments, quick audits	Developers, precise implementation
<i>Comfort & Cybersickness</i>	Broad coverage + empathy focus	Very detailed	Precise rules (20–40 min sessions, snap turning)
<i>Interaction Design</i>	Natural metaphors + multimodal	Embodied focus, avoid micro-movements	Granular feedback (hover highlights, haptics)
<i>UI / Diegetic Design</i>	3D zones + minimal text	Strong diegetic emphasis, cognitive load limits	Angular size, billboard, angular scaling
<i>Visual & Audio Fidelity</i>	Aesthetics + immersion basics	Spatial audio & haptics mentioned	1:1 scale, shadows, 3D audio alignment
<i>Accessibility & Ethics</i>	Inclusive + honest design	Strongest ethics section (addiction, privacy)	Transparent data + diverse abilities
<i>Onboarding & Learnability</i>	Subtle cues + gradual introduction	Contextual tutorials, environmental storytelling	“Teach by doing”, shorten time-to-fun
<i>Testing & Iteration</i>	Early prototyping + user testing	In-headset testing + comfort metrics	Prototype early, in-headset only
<i>Unique Standout Feature</i>	Future trends (AI, haptics, brain interfaces). Personal Favourite	VR-unique framing + assignment tips	Exact metrics
<i>Main Weakness</i>		Some overlap between sections	Can feel too dense for beginners
<i>Tone</i>	Professional	Practical + educational	Technical, guideline-driven

03 COMPREHENSIVE VR/AR UX DESIGN CHECKLIST FROM VIDEOS

1. User-Centred Research and Mindset Shift

Focus on transitioning from 2D to immersive 3D design by prioritizing real user insights and spatial thinking over traditional flat-screen constraints.

- ☐ Conduct user interviews to gather ideas from real-life problem spaces and experiences
- ☐ Create user personas based on research data to represent diverse needs and contexts
- ☐ Map out emotional journeys (story arcs) including introduction, conflict, tension, and resolution
- ☐ Shift mindset from fixed 2D grids/rows/columns to infinite 3D spatial layouts (front, behind, above, below, left, right)
- ☐ Avoid over-relying on trends/hype; base designs on real-world XR experience and testing

2. Spatial Design and Environmental Awareness

Emphasize infinite space, realistic placement, and user context to make environments feel natural and safe.

- ☐ Design for dynamic perspectives where users move and view from multiple angles (unlike fixed 2D viewpoints)
- ☐ Place objects thoughtfully in 3D space (e.g., avoid placing critical content like warnings/errors behind users)
- ☐ Incorporate spatial awareness of physical surroundings (e.g., time of day, device, stress levels, location)
- ☐ Provide safe boundaries, play areas, and options for room-scale, seated, or standing experiences
- ☐ Use natural affordances (hints/tips) to guide users without overwhelming them
- ☐ Walk the scene physically during design to simulate user paths, distances to objects/menus, and ergonomics

3. Interaction and Input Principles

Move from limited 2D inputs (touch, click, swipe) to intuitive, body-based interactions that feel physical and seamless.

- ☐ Design for dynamic perspectives where users move and view from multiple angles (unlike fixed 2D viewpoints)

- ☐ Enable natural, physical interactions like grabbing objects, gaze, gestures, voice commands, or full-body engagement
- ☐ Support first-person or third-person points of view, switching as needed for story flow or emotion
- ☐ Design menus as body-locked (user pulls them out) or world-locked (fixed in environment) based on context
- ☐ Include interactive visualizations (e.g., adjust data, reshuffle) that respond to user actions
- ☐ Make interactions discoverable and frustration-free, avoiding overwhelm in immersive settings

4. Immersion, Feedback, and Sensory Engagement

Enhance presence through multi-sensory cues, making virtual worlds feel real while reducing reliance on visuals alone.

- ☐ Design for dynamic perspectives where users move and view from multiple angles (unlike fixed 2D viewpoints)
- ☐ Optimize first 30 seconds for immediate immersion and "feeling like you're there"
- ☐ Integrate emotions, sound, and movement (ESM) to build tension and response (e.g., spatial audio tilting head naturally)
- ☐ Provide multi-modal feedback: visual (colour changes), audio (spatial sounds), haptic (vibrations via controllers/hands)
- ☐ Use sound and movement together for emotional impact (e.g., running from a threat)
- ☐ Leverage spatial audio to indicate events/directions, mimicking real-life cues

5. Comfort, Motion, and Scale Principles

Prioritize user wellbeing to prevent disorientation, sickness, or fatigue in extended sessions.

- ☐ Ensure realistic 1:1 scale and proportions for objects/environments (e.g., avoid oversized chairs)
- ☐ Maintain steady frame rates (at least 90 FPS) to minimize motion sickness
- ☐ Avoid rapid/involuntary movements; offer multiple locomotion options (teleportation, smooth, world pulling)
- ☐ Respect personal space, especially in multi-user apps (e.g., no unexpected proximity)

- ☐ Incorporate 3D visual hierarchy using size, colour, proximity, contrast, and distance (closer items demand more attention)
- ☐ Make navigation secondary to content delivery, focusing on comfort in head/eye movements

6. Storytelling and Flow Design

Structure experiences like narratives to engage users emotionally and guide them intuitively.

- ☐ Develop story beats (actions/reactions) that respond to user inputs for dynamic flow
- ☐ Use affordance canvas to plan hints, tips, and user perspectives throughout the experience
- ☐ Build story arcs with intro, conflict, tension, and resolution to map emotions
- ☐ Guide users spatially through depth, distance, and immersive environments
- ☐ Change point of view (first/third person) to enhance story beats and emotional shifts

7. Social and Multi-User Design

For collaborative experiences, foster a sense of togetherness through shared elements.

- ☐ Include social signifiers and focus points to bring users together
- ☐ Design shared behaviours, tasks, and environments that create a "being together" feeling
- ☐ Ensure interactions in social UIs feel natural and responsive in virtual spaces

8. Prototyping, Testing, and Iteration

Validate designs early with low-fidelity methods and real users to refine for better results.

- ☐ Start with low-tech mocks (tape, paper, strings) to simulate affordances and flows
- ☐ Test prototypes with 3–5 interviewed users, gathering feedback on comfort and immersion
- ☐ Refine iteratively based on user playtesting throughout development

- ☐ Create UI simulations like video storyboards or improv demos with team members
- ☐ Prototype in tools like Unity/Unreal only after initial research and mocking

03 COMPARISON OF AI-GENERATED CHECKLIST VS. VIDEO CHECKLIST

The comparison focuses on key aspects: **coverage** (what principles are included), **depth/detail** (how granular the items are), **structure** (organization and flow), **similarities** (overlaps in principles), **differences** (unique or missing elements), and **overall strengths**. I've used a table for clarity where effective, followed by a narrative summary.

Key Similarities

Both checklists emphasize user-centred design, spatial thinking, comfort/motion sickness prevention, intuitive interactions, multi-sensory feedback, and iterative prototyping/testing. They align on core XR shifts from 2D (e.g., fixed grids to infinite space, taps/clicks to gestures/gaze). The video-derived one draws from practical anecdotes (e.g., Anna's XR work with Meta/Puma, Peter's user-interview flow), while the AI-generated one incorporates more technical metrics (e.g., FPS thresholds, angular sizes).

Key Differences

- **Scope and Source:** The AI-generated checklist is more exhaustive and platform-agnostic, pulling from broader UX standards (e.g., accessibility customizations, ethical data handling). The video-derived one is narrower, focused on mindset shifts and storytelling, reflecting the videos' emphasis on real-world examples and emotional engagement.
- **Depth:** AI-generated is highly detailed with specific thresholds (e.g., "≥90 FPS", "1.5° FOV for text"); video-derived is more conceptual (e.g., "first 30 seconds for immersion").
- **Length:** AI-generated has ~60 items across 10 sections; video-derived has ~35 items across 8 sections.
- **Unique Focus:** AI-generated stresses technical reliability (e.g., occlusion culling) and future trends (e.g., BCI); video-derived highlights narrative elements (e.g., story arcs, ESM—emotions/sound/movement).

Detailed Comparison Table

This table compares sections and representative items. Not all items are listed for brevity; I've highlighted examples of matches, gaps, and variations.

<i>Aspect</i>	<i>AI-Generated Checklist</i>	<i>Video Checklist</i>	<i>Comparison Notes</i>
<i>User Research & Mindset</i>	Thorough research, empathy, personas, inclusivity	Interviews, personas, emotional mapping; 2D-to-spatial shift	High overlap: video emphasizes emotional arcs; AI adds accessibility
<i>Spatial Design & Awareness</i>	Detailed zones/FOV, scale, landmarks	Infinite space, thoughtful placement, amplified awareness	Similar core; AI more metric-based; video more cautionary
<i>Interactions & Inputs</i>	Multimodal, direct/indirect, avoid micro-movements	Physical (grab/gaze), body/world-locked menus	Strong match; video links to storytelling; AI more specific
<i>Immersion & Feedback</i>	Multi-modal (visual/audio/haptic), spatial audio	ESM (emotions/sound/movement), haptic	Overlap on senses; video unique in emotional ESM
<i>Comfort & Motion</i>	FPS/latency thresholds, locomotion, breaks	90 FPS, multiple locomotion, personal space	Near identical, AI adds timers; video stresses multi-user
<i>Storytelling & Flow</i>	Gradual onboarding, environmental cues	Story arcs/beats, affordance canvas	Video stronger on narrative; AI integrates into onboarding
<i>Social/Multi-User</i>	Indirect via safety	Dedicated: signifiers, shared tasks, togetherness	Video more dedicated; AI covers indirectly
<i>Prototyping & Testing</i>	In-headset, iteration, low-tech mocks	Low-tech (tape/paper), 3–5 users, improv	Excellent alignment: both include "walk the scene"
<i>Ethics & Futureproofing</i>	Data protection, emerging trends	Implied in comfort	AI unique in ethics/future aspects
<i>UI Presentation</i>	Diegetic UI, hierarchy, minimize text	Hierarchy via distance	AI more detailed; video integrates into spatial